

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

Medical image analysis requires robust visual representations despite the persistent scarcity and cost of expert annotation. Self-supervised masked image modeling (MIM) directly addresses this constraint, yet prevailing methods still rely primarily on random, high-ratio masking, which is computationally inefficient and only weakly aligned with task-relevant anatomy or pathology. Existing region-aware variants partially mitigate this limitation but typically depend on reconstruction-error heuristics or supervised localization signals, restricting their flexibility across modalities and downstream objectives. We introduce Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. The framework uses vision-language models to obtain prompt-conditioned localization cues and then applies differentiated masking so that diagnostically relevant regions receive focused reconstruction pressure while semantically redundant background is masked more lightly. This design converts open-vocabulary textual guidance into a pretraining signal that improves semantic alignment, representation quality, and transfer across the evaluated medical imaging tasks. Across brain MRI, chest CT, and lung X-ray experiments, Mask What Matters consistently improves over existing MIM baselines, including SparK, with gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection and instance segmentation evaluation. These gains are obtained with substantially lower overall masking ratios, such as 40% for MWM versus 70% for SparK in the masking-ratio comparison. The results establish controllable, text-driven masking as an effective mechanism for semantically aligned self-supervised learning in medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection constitute the core predictive tasks through which computer vision systems support medical image analysis. Deep learning has substantially advanced these tasks, but its strongest empirical performance usually depends on large annotated datasets. In medical imaging, where expert labels are expensive, time-consuming, and often difficult to standardize across modalities, this dependence remains a central obstacle to scalable representation learning.

Self-supervised learning (SSL) addresses this bottleneck by pretraining models on unlabeled data and transferring the resulting representations to annotated downstream tasks [Jaiswal et al. \(2020\)](#); [Liu et al. \(2021\)](#); [Krishnan et al. \(2022\)](#); [Huang et al. \(2023\)](#). Among SSL paradigms, masked image modeling (MIM) has become a particularly influential generative approach: by hiding image regions and requiring reconstruction from visible context, MIM encourages encoders to capture structural regularities and semantic dependencies. This paradigm has achieved strong transfer performance in natural-image settings and has been adapted successfully to medical image analysis [He et al. \(2022\)](#); [Gupta et al. \(2025\)](#); [Xiao et al. \(2023\)](#).

The dominant MIM formulations, including masked autoencoders (MAE) [He et al. \(2022\)](#) and SparK [Tian et al. \(2023\)](#), nevertheless mask image content largely at random. Because random

masking is agnostic to the regions that define downstream objectives, it creates two linked limitations. First, the reconstruction task may allocate substantial capacity to visual content that is not semantically decisive for transfer. Second, to increase the probability that informative regions are hidden, MIM methods often adopt very high masking ratios, such as 75% in MAE, thereby increasing reconstruction difficulty and resource demand. This contrasts with masked language modeling, where the compact and semantically dense structure of text enables effective learning under lower masking rates (Devlin et al., 2019). The need for high-ratio masking in images therefore reflects not only visual redundancy but also the absence of a mechanism that selectively emphasizes the regions that matter.

This mismatch is especially consequential in medical imaging. Diagnostic evidence is frequently spatially sparse, appearing in localized lesions, organs, or anatomical structures, whereas large portions of an image may be clinically redundant for a given task. At the same time, medical imaging spans MRI, CT, and X-ray modalities whose task-relevant regions differ markedly in shape, scale, and spatial distribution. A useful masking strategy must therefore be both selective enough to align pretraining with the downstream semantic target and flexible enough to operate across heterogeneous imaging conditions.

We address these challenges with a controllable, text-guided masking framework, **Mask-What-Matters (MWM)**. Given a user-specified open-vocabulary phrase or description, a pretrained vision-language model provides zero-shot localization cues that are converted into robust regions of interest (ROIs). MWM then performs prompt-conditioned, region-specific masking: semantically important areas, such as lesions or organs, receive higher masking pressure, whereas background regions are masked at lower ratios. In this way, MWM injects downstream semantic intent into self-supervised pretraining without requiring per-image reports or labels.

Our contributions are threefold:

- We formulate controllable text-guided masking for medical image MIM. MWM integrates vision-language localization (BiomedCLIP) with segmentation refinement (SAM) to derive task-relevant regions from open-vocabulary prompts and to apply differentiated ROI versus background masking, directly addressing the semantic misalignment of random masking.
- We provide an annotation-free and backbone-agnostic pretraining design. MWM requires no per-image reports or labels and can be integrated into both ViT- and ConvNet-based MIM pipelines under a unified protocol.
- We demonstrate consistent empirical gains across modalities and tasks. On brain MRI, chest CT, and X-ray datasets, MWM outperforms representative MIM baselines, including SparK, improving classification, detection, and segmentation performance within the evaluated settings.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling was first established in natural language processing by bidirectional encoder representations from transformers (BERT) (Devlin et al., 2019), which learns contextual representations by masking tokens and predicting them during pretraining. He et al. (2022) transferred this principle to computer vision through MIM, in which image regions are hidden and reconstructed from the remaining visual context. The resulting paradigm has shown strong generalization in both natural and medical image domains (Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023). Representative methods include MAE (He et al., 2022), MedMAE (Gupta et al., 2025; Xiao et al., 2023), and SparK (Tian et al., 2023): MAE uses high-ratio random masking to train ViTs toward higher-level semantic representations; MedMAE adapts the paradigm to medical images; and SparK extends MIM to convolutional backbones by exploiting multi-scale feature hierarchies.

Despite this progress, conventional MIM typically remains random or structure-agnostic at the masking stage and therefore does not explicitly encode the semantic requirements of downstream tasks. Non-selective masking can lead pretraining to overemphasize irrelevant regions and reduce transfer effectiveness. Recent region-aware methods seek to improve this alignment, with two main directions emerging.

The first direction uses loss-driven adaptive masking. Methods such as hard patches mining (HPM) [Wang et al., (2023a)] and AnatoMask [Li et al., (2024)] analyze regional reconstruction errors and assign higher masking probabilities to high-loss areas, with the objective of prioritizing semantically valuable structures. This criterion, however, can conflate reconstruction difficulty with task relevance. In brain magnetic resonance imaging (MRI), for example, complex structures such as the spinal cord or eyeballs may attract masking attention even when they are unrelated to tumor-oriented downstream tasks.

The second direction incorporates external perceptual modules. FocusMAE [Basu et al., (2024)], for instance, uses pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Although such modules can be effective in specific tasks, they require supervised RPN training on domain-specific annotated data, which limits flexibility when tasks or modalities change.

These limitations motivate a masking mechanism that is semantically aligned, flexible across tasks, and not dependent on per-image supervision. Natural language provides precisely this kind of compact and interpretable guidance, making it well suited for task-aware MIM when it can be linked reliably to image regions.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Recent progress in vision-language alignment has made zero-shot and open-vocabulary localization increasingly practical [Radford et al., (2021)]. The Contrastive Language-Image Pretraining (CLIP) family learns a shared embedding space for images and text through large-scale image-text pretraining and has become a foundation for zero-shot and open-set visual recognition. In medical imaging, BiomedCLIP [Zhang et al., (2023)] strengthens cross-modal understanding for biomedical content and supports robust performance across downstream applications. Most uses of vision-language pretrained models nevertheless remain concentrated in supervised downstream tasks, including object detection and semantic segmentation [Zhong et al., (2022), Aleem et al., (2024), Koleilat et al., (2024)]. Their open-vocabulary and zero-shot capabilities have not been fully exploited as mechanisms for controlling the MIM pretraining process itself.

Taken together, prior region-aware MIM methods improve semantic focus but remain constrained by additional supervision, imperfect alignment between masking criteria and task semantics, or limited adaptability. At the same time, vision-language models offer a direct route from textual task intent to visual localization. MWM connects these two lines of work by using natural language prompts to guide where MIM should allocate its reconstruction pressure during pretraining.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** consists of two coupled stages: text-guided region localization and region-aware masked image modeling. The localization stage uses a frozen vision-language model to map an open-vocabulary prompt to task-relevant image regions. The pretraining stage then assigns differentiated masking ratios to those regions and to the remaining background, making the reconstruction objective explicitly responsive to the semantic target specified by the prompt.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM obtains task-relevant regions through a structured localization pipeline. It first constructs textual prompts and uses BiomedCLIP to embed the prompt and image in a shared cross-modal space. A Multi-Modal Information Bottleneck (M2IB) module then produces a prompt-conditioned saliency map, which is binarized with K-Means and refined by the Segment Anything Model (SAM). The refined mask is finally converted into a bounding box with a controllable expansion margin, yielding the spatial guide used by the masked modeling stage.

Prompt Generation. Textual prompts define the semantic target of localization and therefore determine which image regions are emphasized during pretraining. To obtain descriptive guidance, large

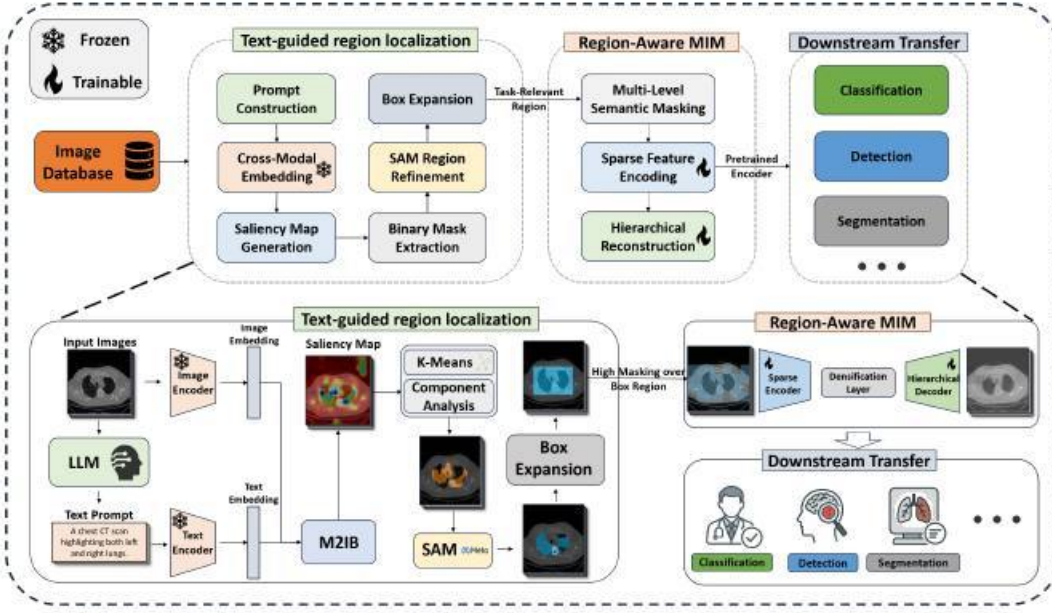


Figure 1: Overview of the MWM framework. The top panel presents the complete workflow: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel details how open-vocabulary prompts are transformed into localization cues and differentiated masking decisions.

language models (LLMs) are queried with structured instructions such as “Describe the typical visual characteristics of a [category] in a [modality] image”. We also evaluate simple category phrases as alternative prompts, allowing the experiments to compare concise and descriptive language under the same localization and pretraining pipeline.

Cross-Modal Embedding. Given a prompt, BiomedCLIP [Zhang et al. (2023)] encodes the medical image and text into a shared semantic space. For an image I and prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These embeddings provide the cross-modal representation on which prompt-conditioned saliency estimation is based.

Saliency Map Generation. A cross-modal saliency module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. (2023b)] localizes task-relevant visual evidence by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns an importance weight to each spatial location according to its semantic relevance to the prompt. M2IB optimizes the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the BiomedCLIP image and text embeddings, respectively, and γ controls the trade-off between preserving cross-modal relevance and filtering task-irrelevant visual information.

Binary Mask Extraction. The saliency map is binarized with unsupervised K-Means clustering [Lloyd (1982)] to produce an initial foreground mask guided by the prompt. The map is partitioned into two pixel groups, and the group with higher saliency values is treated as foreground. Connected component analysis is then applied to the binary map, retaining the largest foreground regions and suppressing small false activations, especially along noisy boundaries.

Region Refinement. SAM is used as a spatial refinement module to improve the geometric fidelity of the localized region [Kirillov et al. (2023)]. For each selected connected region $c_i \in \mathcal{C}^*$, the minimal enclosing bounding box is computed and supplied as the visual prompt for SAM.

Given the original image I and the corresponding bounding boxes, SAM predicts the refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} preserves a more faithful spatial structure than the coarse saliency-derived region.

Box-Based Region Expansion. Because the SAM output may still contain boundary irregularities, directly using the mask can introduce unstable spatial guidance during representation learning. MWM therefore converts the refined SAM mask into a bounding box with a controllable expansion margin. This expanded box defines the final region of interest (ROI) for the masked modeling procedure.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the ROI produced by the localization module in Section 3.2, MWM performs masked image modeling through three operations: multi-level semantic masking, sparse feature encoding, and hierarchical reconstruction. The masking operation assigns different ratios according to region importance, the encoder processes only visible patches, and the decoder reconstructs the image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM uses the localized semantic regions to define a multi-level masking pattern. Patches inside prompt-highlighted regions, such as tumor- or organ-related areas, are masked at a higher ratio so that reconstruction pressure concentrates on task-relevant structures. Background patches are masked lightly or left visible, reducing redundant reconstruction of low-value image content. Figure 2 visualizes the resulting masking behavior.

Sparse Feature Encoding. Visible patches are assembled into a sparse image and encoded following the SparK design [Tian et al., 2023]:

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i denotes the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ denotes the resulting multi-scale features. Although our implementation instantiates MWM with convolutional backbones, the same interface applies to ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image through a top-down feature hierarchy. At each scale l , masked locations are densified with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , which is then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

The reconstruction objective is the mean-squared error over masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i is the original patch, and $\hat{x}_i$ is its reconstruction.

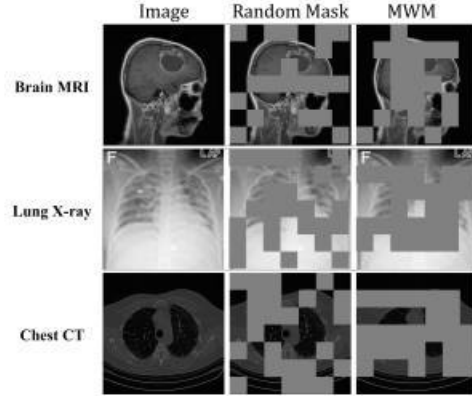


Figure 2: Comparison of masking strategies. Gray blocks denote masked regions.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Three datasets spanning different imaging modalities are used for self-supervised pretraining. (1) *Brain MRI* includes approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering various tumor types and anatomical structures. (2) *Lung X-ray* consists of around 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021)]; Chowdhury et al. (2020), including COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* comprises 12,000 slices from a lung disease dataset [Konya (2020)], with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Three datasets are used for downstream evaluation across classification and detection tasks. (1) *Brain Tumor MRI* [Feltrin (2023)] includes 4,479 MRI images spanning 44 tumor types, such as astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermary et al. (2018)] consists of 5,863 pediatric chest X-rays categorized as either normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] includes CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods use backbones of similar model scale, with convolution-based methods adopting ResNet-50 [He et al. (2016)] and transformer-based methods using ViT-S [Dosovitskiy et al. (2020)]. The input resolution is 224×224 , and models are initialized with ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} . Downstream fine-tuning retains the same encoder and input resolution. All experiments are performed on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, we fix non-target variables (e.g., prompt type or SAM usage) to their empirically optimal settings during evaluation to ensure controlled comparison.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first evaluate the effectiveness of the text-guided localization module across three imaging modalities—brain MRI, lung X-ray, and chest CT—and further examine the impact of prompt type and SAM refinement on localization performance. Specifically, we compare the predicted bounding boxes with expert-annotated segmentation masks, and report occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

As shown in Table I, our text-guided localization with effective prompts achieves consistently high occlusion recall across all datasets (around 0.82), while maintaining reasonable precision. This

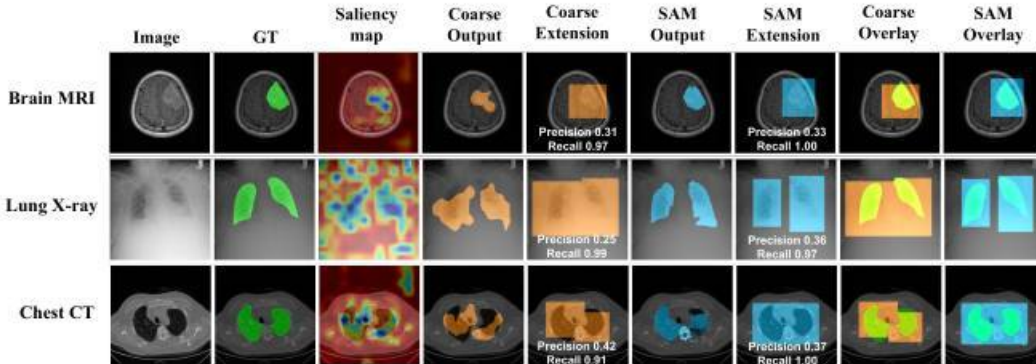


Figure 3: Representative text-guided localization results across medical imaging modalities.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance (Occlusion Precision/Recall) across prompt types and refinement strategies. *Coarse Ext.* denotes bounding box extension of the coarse region; + *SAM Ext.* indicates further refinement using SAM.

confirms the framework’s ability to identify task-relevant regions solely from textual descriptions, despite modality-specific anatomical variations (see Figure 3 for visualized examples).

Effect of Prompt Type on Region Localization. To further assess the impact of language design, we evaluate the effect of two prompt types on localization performance. As shown in Table 1, descriptive sentence prompts yield better results in Brain MRI and Chest CT, indicating that richer semantics aid localization. In contrast, concise phrases outperform in lung radiographs in terms of recall (0.82 vs. 0.66), likely because the lungs’ broad anatomical coverage allows general terms to be sufficiently effective.

Effect of SAM Refinement on Region Localizatio. We also evaluate the contribution of spatial refinement by applying SAM to the coarse masks. As shown in Table 1, SAM refinement improves precision while maintaining high recall. For instance, in Brain MRI, precision increases from 0.22 to 0.30, while recall remains above 0.94. This demonstrates the utility of SAM in refining coarse localization outputs.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We then evaluate the generalization and representation ability of MWM by comparing it with representative self-supervised learning (SSL) methods across both reconstruction-based and contrastive learning paradigms. All methods are pretrained and evaluated using consistent datasets and training settings to ensure a fair comparison.

Image Classification. Classification performance on three downstream medical imaging datasets is examined. Following common practice, results are reported under two settings: (1) *full fine-tuning*, where the entire model is updated; and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained on top.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM and several representative self-supervised methods, including MAE [He et al. (2022)], SparK [Tian et al. (2023)],

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) on downstream datasets, where MIM stands for Masked Image Modeling, while CL stands for Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets (frozen encoder).

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) results (%) on the BR35H dataset.

AnatoMask [Li et al. \(2024\)](#), SimCLR [Chen et al. \(2020a\)](#), MoCoV2 [Chen et al. \(2020b\)](#), and BYOL [Grill et al. \(2020\)](#). MWM consistently achieves the highest performance, demonstrating strong generalization across diverse imaging modalities. The most significant gain is observed on the Chest CT dataset, where MWM achieves 97.5% accuracy—surpassing SparK by +3.1 and AnatoMask by +1.7. This improvement highlights the benefit of text-guided masking, which emphasizes semantically critical yet spatially sparse regions—such as lung lobes in CT scans—while avoiding masking redundancy.

- **Linear probing:** To further evaluate the quality of learned representations, linear probing experiments are conducted. For simplicity, MWM is compared with two recent state-of-the-art masked image modeling methods—SparK and AnatoMask. As shown in Table 3, MWM significantly outperforms both across all datasets, highlighting its ability to learn effective features through text-driven pretraining.

Object Detection and Instance Segmentation. To assess the generalizability of learned representations beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset [Hamada \(2025\)](#). As shown in Table 4, MWM achieves the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), surpassing SparK by +1.3 and +1.1 points, respectively. These results demonstrate that MWM transfers effectively to both object-level localization and pixel-wise segmentation.

In summary, incorporating text-driven masking during pretraining enables the model to learn more informative and transferable representations. This semantic guidance consistently improves performance across classification, detection, and segmentation tasks, underscoring its effectiveness in enhancing visual representation learning.

4.5 ABLATION STUDY

We also conduct ablation experiments to evaluate the individual impact of prompt design, SAM-based region refinement, and masking ratio on downstream classification performance in MWM.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, while the opposite trend is observed on lung X-ray. This aligns with the localization results in Section 4.3, where prompts that enabled more accurate region identification also led to better downstream results. Importantly, both types of prompts—regardless of granularity—consistently outperform random masking across all datasets. This confirms that text-guided masking enables effective semantic alignment between pretraining and downstream tasks, which is crucial for improving self-supervised learning performance. In addition, an ablation on SAM demonstrates that incorporating this module during pretraining consistently improves downstream classification, confirming its effectiveness in enhancing text-guided masking.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under different prompt types and region refinement strategies across three imaging modalities. *No Prompt* refers to random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM achieves 96.8% classification accuracy at a masking ratio of just 40%—a setting substantially lower than around 70% typically used in masked image modeling. Despite the reduced ratio, it outperforms SparK’s best performance at 70% masking by +2.4% in absolute accuracy. To rule out the possibility that SparK’s inferior performance is due to suboptimal masking configurations, SparK under 40%, 50%, 60%, and 70% masking ratios are further evaluated. The results confirm that 70% masking yields the highest accuracy for SparK, validating the robustness of our comparison. These results challenge the prevailing assumption that high masking ratios are inherently optimal for masked image modeling. Instead, they highlight the advantage of semantically guided, task-aware masking strategies like MWM, which achieve stronger downstream performance with lower masking ratios by preserving task-relevant content and reducing redundancy.

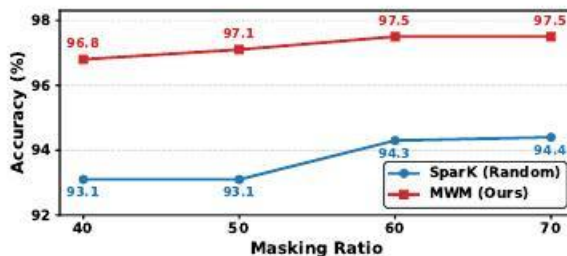


Figure 4: Classification performance on the Chest CT dataset under different masking strategies and ratios.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that integrates text-guided semantic localization with region-aware masking. By leveraging natural language prompts to highlight task-relevant areas and applying differentiated masking, MWM enables semantically aligned representation learning without requiring per-image annotations.

Comprehensive experiments demonstrate the effectiveness and generalizability of MWM: (1) natural language prompts reliably guide semantic region localization, validating the feasibility of text-driven masking; (2) MWM consistently outperforms existing methods (e.g., SparK, AnatoMask) in classification across three imaging modalities, and also yields gains in detection and instance segmentation; (3) MWM maintains strong performance even at lower masking ratios (e.g., 40%), underscoring the benefit of semantic guidance in self-supervised pretraining.

Limitations. While promising, our framework has several limitations. First, although text prompts offer flexible guidance, the robustness of MWM to variations in prompt style, vocabulary, or noise has not been systematically examined. Second, the reliance on external textual descriptions introduces a dependency that may weaken generalization in scenarios where reliable prompts are unavailable or inconsistent.

Looking forward, we believe that text-driven, region-aware masking offers a principled path toward more semantically grounded self-supervised learning, and that extending MWM beyond medical imaging may open new opportunities for broader vision–language pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

We used LLMs *only* for language editing and presentation polishing (e.g., grammar, phrasing, and minor stylistic clarity) of text written by the authors. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.